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Computational Logic

1- Semantics of Propositional Logic - Solutions

(lab classes 1 and 2)

1.1) For each of the following sentences, say whether it is a proposition. If so, state whether it is true or false.

(a) Porto is the capital of Portugal.

(b) In 1910 the Portuguese Republic was proclaimed.

(c) Shall I compare thee to a summer's day.

(d) $1+2+3$

(e) $1+2+3=6$

(f) 7 is an even number.

(g) $x+y$ is an even number.

(h) There is an x such that $7+x$ is an even number.

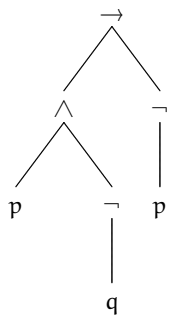
Solution: A proposition is a statement that is either true or false.

	Proposition	T/F
(a)	✓	**F**
(b)	✓	**T**
(c)	×	-
(d)	×	-
(e)	✓	**T**
(f)	✓	**F**
(g)	×	-
(h)	✓	**T**

In fact (h) is a proposition in first order logic.

1.3) Considering the priority conventions of logical operations given in the course ($\neg < \wedge < \vee < \rightarrow$ and $\rightarrow$ right associative), for each of the following formulas draw its syntactic tree and indicate its subformulas:
Solutions:

(a) $p \wedge \neg q \rightarrow \neg p$;



Subformulas:

$p \wedge \neg q$

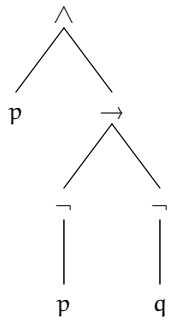
$\neg p$

p

$\neg q$

q

(b) $p \wedge (\neg q \rightarrow \neg p)$;



Subformulas:

p

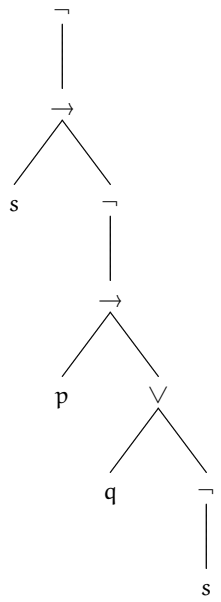
$\neg q \rightarrow \neg p$

$\neg q$

$\neg p$

q

(c) $\neg(s \rightarrow (\neg(p \rightarrow (q \vee \neg s))))$;



Subformulas:

$s \rightarrow (\neg(p \rightarrow (q \vee \neg s)))$

s

$\neg(p \rightarrow (q \vee \neg s))$

$p \rightarrow (q \vee \neg s)$

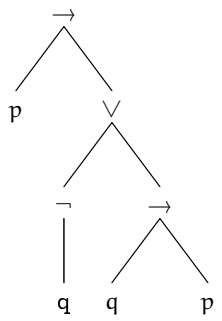
p

$q \vee \neg s$

q

$\neg s$

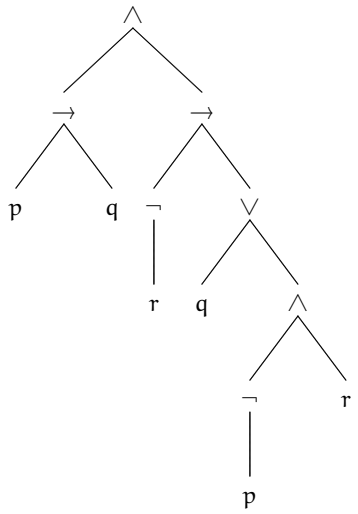
(d) $p \rightarrow (\neg q \vee (q \rightarrow p))$;



Subformulas:

p
$\neg q \vee (q \to p)$
$\neg q$
q
$q \to p$

(e) $(p \to q) \wedge (\neg r \to (q \vee (\neg p \wedge r)))$.



Subformulas:

$p \to q$
p
q
$\neg r \to (q \vee (\neg p \wedge r))$
$\neg r$
r
$q \vee (\neg p \wedge r)$
$\neg p \wedge r$
$\neg p$

1.4) Construct truth tables for each of the following propositions.
Solutions: We illustrate different ways to present truth tables.

(a) $p \to p$

p	$\to$	p
T	**T**	**T**
T	**T**	**T**
F	**T**	**F**
F	**T**	**F**

The formula is therefore a tautology.

(b) $\neg q \wedge (p \to q)$

p	q	$\neg$	q	$\wedge$	$(p$	$\to$	$q)$
T	**T**	**F**		**F**		**T**	
T	**F**	**T**		**F**		**F**	
F	**T**	**F**		**F**		**T**	
F	**F**	**T**		**T**		**T**	

The formula is therefore satisfiable.

(c) $(p \wedge q) \rightarrow \neg(p \vee q)$

p	q	$p \wedge q$	$p \vee q$	$\neg(p \vee q)$	$(p \wedge q) \rightarrow \neg(p \vee q)$
T	**T**	**T**	**T**	**F**	**F**
T	**F**	**F**	**T**	**F**	**T**
F	**T**	**F**	**T**	**F**	**T**
F	**F**	**F**	**F**	**T**	**T**

The formula is therefore satisfiable.

(d) $\neg(p \vee q) \rightarrow (\neg p \wedge \neg q)$

$\neg$	$(p$	$\vee$	$q)$	$\rightarrow$	$(\neg$	p	$\wedge$	$\neg$	$q)$
F	**T**	**T**	**T**	**T**	**F**	**T**	**F**	**F**	**T**
F	**T**	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**
F	**F**	**T**	**T**	**T**	**T**	**F**	**F**	**F**	**T**
T	**F**	**F**	**F**	**T**	**T**	**F**	**T**	**T**	**F**

The formula is therefore a tautology.

1.5) When $\nu(p) = \mathbf{T}$ and $\nu(q) = \mathbf{F}$ determine if $\models_\nu \phi$, where ϕ is:

(a) $(p \vee q) \rightarrow (p \wedge q)$;
Solution: From $\nu(p) = \mathbf{T}$ it follows that $\models_\nu p$, therefore $\models_\nu p \vee q$. On the other hand $\nu(q) = \mathbf{F}$ implies that $\not\models_\nu q$, therefore $\not\models_\nu p \wedge q$. Therefore $\not\models_\nu (p \vee q) \rightarrow (p \wedge q)$.

(b) $(p \vee \neg q) \wedge (\neg p \wedge q)$;
Solution: Since $\nu(q) = \mathbf{F}$ then $\not\models_\nu q$. This implies that $\not\models \neg p \wedge q$, from which follows that $\not\models_\nu (p \vee \neg q) \wedge (\neg p \wedge q)$.

(c) $(p \rightarrow q) \wedge (p \rightarrow \neg q)$;
Solution: From $\nu(p) = \mathbf{T}$ and $\nu(q) = \mathbf{F}$ it follows that $\models_\nu p$ and $\not\models_\nu q$, therefore $\not\models_\nu p \rightarrow q$, which implies $\not\models_\nu (p \rightarrow q) \wedge (p \rightarrow \neg q)$.

(d) $p \rightarrow (q \rightarrow (p \vee q))$;
Solution: From $\nu(q) = \mathbf{F}$ it follows that $\not\models_\nu q$, therefore $\models_\nu q \rightarrow (p \vee q))$, which in turn implies $\models_\nu p \rightarrow (q \rightarrow (p \vee q))$.

(e) $((p \rightarrow q) \rightarrow p) \rightarrow p$;
Solution: From $\nu(p) = \mathbf{T}$ it follows that $\models_\nu p$, therefore $\models_\nu ((p \rightarrow q) \rightarrow p) \rightarrow p$.

1.8) Which of the following statements are valid? Justify.

(a) $(p \rightarrow q) \wedge (p \rightarrow r) \Leftrightarrow (p \rightarrow (q \wedge r))$
Solution: There are many ways to show that the statement is valid. We can prove the statement by a sequence of logic equivalences using the equivalence laws of propositional logic.

$$\begin{array}{lll} (p \rightarrow q) \wedge (p \rightarrow r) & \Leftrightarrow & (\neg p \vee q) \wedge (\neg p \vee r) \quad (\text{by } \phi \rightarrow \psi \Leftrightarrow \neg\phi \ \vee \ \psi) \\ & \Leftrightarrow & \neg p \vee (q \wedge r) \quad (\text{using distributivity}) \\ & \Leftrightarrow & (p \rightarrow (q \wedge r)) \quad (\text{again, by } \phi \rightarrow \psi \Leftrightarrow \neg\phi \ \vee \ \psi) \end{array}$$

Also note that $\varphi \Leftrightarrow \psi$ iff $\varphi \models \psi$ and $\psi \models \varphi$. Therefore the formulas are equivalents if every assignment that satisfies φ also satisfies ψ and vice-versa. This can be verified by checking that the truth tables of both formulas coincide.

$(p$	$\rightarrow$	$q)$	$\wedge$	$(p$	$\rightarrow$	$r)$
T	**T**	**T**	**T**	**T**	**T**	**T**
T	**T**	**T**	**F**	**T**	**F**	**F**
T	**F**	**F**	**F**	**T**	**T**	**T**
T	**F**	**F**	**F**	**T**	**F**	**F**
F	**T**	**T**	**T**	**F**	**T**	**T**
F	**T**	**T**	**T**	**F**	**T**	**F**
F	**T**	**F**	**T**	**F**	**T**	**T**
F	**T**	**F**	**T**	**F**	**T**	**F**

p	$\rightarrow$	$(q$	$\wedge$	$r)$
T	**T**	**T**	**T**	**T**
T	**F**	**T**	**F**	**F**
T	**F**	**F**	**F**	**T**
T	**F**	**F**	**F**	**F**
F	**T**	**T**	**T**	**T**
F	**T**	**T**	**F**	**F**
F	**T**	**F**	**F**	**T**
F	**T**	**F**	**F**	**F**

(b) $(p \rightarrow r) \wedge (q \rightarrow r) \Leftrightarrow ((p \wedge q) \rightarrow r)$
Solution: This statement is not valid. Again this can be verified using truth tables to find an assignment for which the formulas have different truth values. But it is easy to see that, for $\nu(p) = \mathbf{T}$, $\nu(q) = \mathbf{F}$ and $\nu(r) = \mathbf{F}$, then $\not\models_\nu (p \rightarrow r) \wedge (q \rightarrow r)$ because $\not\models_\nu p \rightarrow r$, and $\models_\nu (p \wedge q) \rightarrow r$, since $\not\models_\nu p \wedge q$.

(c) $(p \wedge q) \vee r \Leftrightarrow (p \vee r) \wedge (q \vee r)$
Solution: This statement is the equivalence law for the distributivity of $\vee$ over $\wedge$, therefore it is valid. To prove a basic equivalence law, one uses truth tables.

$(p$	$\wedge$	$q)$	$\vee$	r
T	**T**	**T**	**T**	**T**
T	**T**	**T**	**T**	**F**
T	**F**	**F**	**T**	**T**
T	**F**	**F**	**F**	**F**
F	**F**	**T**	**T**	**T**
F	**F**	**T**	**F**	**F**
F	**F**	**F**	**T**	**T**
F	**F**	**F**	**F**	**F**

$(p$	$\vee$	$r)$	$\wedge$	$(q$	$\vee$	$r)$
T	**T**	**T**	**T**	**T**	**T**	**T**
T	**T**	**F**	**T**	**T**	**T**	**F**
T	**T**	**T**	**T**	**F**	**T**	**T**
T	**T**	**F**	**F**	**F**	**F**	**F**
F	**T**	**T**	**T**	**T**	**T**	**T**
F	**F**	**F**	**F**	**T**	**T**	**F**
F	**T**	**T**	**T**	**F**	**T**	**T**
F	**F**	**F**	**F**	**F**	**F**	**F**

1.10) Imagine an island where some of its inhabitants, called "knights", always tell the truth, and others called "knaves" always lie. We assume that any inhabitant of this island falls into one or the other class.

(a) Two inhabitants, **A** and **B** are in a garden. **A** makes the following statement: "At least one of us is a knave." What are **A** and **B**?
Solution: **A** cannot be a knave and say that at least one of them is a knave, otherwise he would be stating the truth. Therefore, **A** is a knight, which implies that **B** is a knave.

(b) Two inhabitants, **A** and **B** are in a garden. Suppose A says, "I am a knave, or B is a knight." What are **A** and **B**?
Solution: Once again **A** cannot be a knave and make this statement. Therefore, **A** is a knight, and **B** must also be a knight, otherwise the statement would be false.

(c) Two inhabitants, **A** and **B**, are in a garden. Suppose **A** says, "I am a knave, or two plus two equals five." What can we conclude?
Solution: **A** cannot make such statement, regardless of being a knight or a knave. If **A** were a knave then its statement would be truth, which cannot happen since knaves always lie. If **A** were a knight then the statement would be false, which cannot happen since knights always tell the truth.

(d) Two inhabitants of the island, **A** and **B**, were together in a garden. Suppose **A** says, "I am a knave, but **B** is not." What are **A** and **B**?
Solution: Once again **A** cannot be a knight and make this statement. Therefore **A** must me a knave, in which case **B** must also be a knave, otherwise **A** would be a knave telling the truth.

(e) Three townspeople, **A**, **B**, and **C** come together. **A** and **B** make the following statements:

A: We are all knaves.

B: Exactly one of us is a knight.

What are **A**, **B**, and **C**?
Solution: **A** cannot be a knight and make this statement. Therefore **A** is a knave, which implies that either **B** or **C** is a knight. Note that, they cannot both be knights otherwise **B**'s statement would not be true. And it cannot be the case of **B** being a knave and **C** being a knight, otherwise **B** would be a knave telling the truth. Therefore **B** is a knight and **C** is a knave.

(f) Suppose that in the previous problem, **A** and **B** say the following:

A: We are all knaves.

B: Exactly one of us is a knave.

Can we determine what **B** is? Can we determine what **C** is?
Solution: Again, **A** cannot be a knight and make this statement. Therefore **A** is a knave, which implies that either **B** or **C** or both are knights. They can both be knights, since in this case **A** would be the only knave and **B**'s statement is still true. **B** can also se a knave and **C** a knight, in which case **B**'s statement is false. **B** cannot be a knight and **C** as knave, otherwise he would be a knight telling a lie.

(g) We have three inhabitants, **A**, **B**, and **C** (knights or knaves). **A** says: "**B** and **C** are of the same kind." Someone then asks **C**: "**A** and **B** are of the same kind?" What did **C** say?
Solution: **C** always answers "yes". If **A** is a knight, then either both **B** and **C** are knights, in which case **C** will answer truthfully "yes", or **B** and **C** are knaves, in which case **C** will be lying when answering "yes". If **A** is a knave, then either **B** is a knave and **C** a knight, in which case **C** will answer truthfully "yes", or **B** is a knight and **C** a knave, in which case **C** will be lying when answering "yes".

(h) Once again we are faced with three inhabitants, **A**, **B**, and **C**. Two of them are of the same type (knights or knaves). **A** and **B** make the following statements:

A: **B** is a knave.

B: **A** and **C** are of the same type.

What is **C**?
Solution: **C** is a knave. It cannot be the case that **A** and **B** are of the same type: if they were both knights then **A** would be lying, contradicting the fact of being a knight. If they were both knaves, then **A** would be telling the truth, contradicting the fact of him being a knave. Also it cannot be the case that **A** and **C** are both knights, since in this case **B** is a knave answering truthfully that **A** and **C** are of the same type. If **A** and **C** are both knaves, then **B** will be a knight stating that exactly. **B** and **C** cannot be both knights, because thern

(i) Three of the island's inhabitants, **A**, **B**, and **C**, were together in a garden. A stranger who was passing by asked **A**, "Are you a knight or a knave?" **A** answered, but in such a way that the stranger could not understand the answer. The foreigner then asked **B**, "What did **A** say?" **B** replied: "**A** said he was a knave." At that point **C** exclaimed: "Don't believe **B**, he's lying!" What are **B** and **C**?

Solution: It is impossible for either a knight or a knave to say *"I am a knave"*, since a knight would be lying by saying that, and a knave would be telling the truth. Therefore, **B** is lying, which means that he is a knave. And by point that out, **C** is a knight.

(j) On that island there is a fork in the road, one of the paths leading to the nearest city. If a foreigner wants to know which is the right path, what question should he ask an inhabitant at the fork, in such a way that he can answer with a simple "yes" or "no". **Suggestion:** The question should relate (by equivalence) what you want to know with the determination of the class of the inhabitant.
Solution: The question could be: *"What would you answer if I asked you: 'does the path on the right leads to the nearest city?' "*. If the path on the right leads to the nearest city, then a knight would say that he would answer *"yes"*, because he would be telling the truth. Since the knave would answer *"no"* to such a question, then in order for him to lie to the main question, he would also say *"yes"*. If the path on the right does not lead to the nearest city, then a knight would say that he would answer *"no"*. Since the knave would answer *"yes"* to such a question, then in order for him to lie to the main question, he would also say *"no"*.

(k) **An adventure of mine** This is an unusual problem, and besides it is inspired by a real case. Once when on the island of knights and knaves, I found two inhabitants resting under a tree. I approached and asked one of them: "Is either of you a knight?". He answered, and with that I learned the answer to my question.
Was the person I questioned a knight or a knave? And the other? I can assure you that I gave the necessary information to solve the problem.
Solution: If the person had answered *"yes"* then there is no way I would have known the answer to my question (it could be a knight telling the truth or a knave lying), therefore the answer must have been *"no"*, in which case the person cannot be a knight. Therefore the person that answered is a knave and by saying *"no"*, it means that the other person is a knight.

(l) Suppose you visited the island of knights and knaves and found two of its inhabitants lazily resting in the sun. You ask one of them if the other is a knight and you receive the answer (yes or no). Then you ask the other if the first is a knight and you receive the respective answer (yes or no).
Are the two answers necessarily the same?
Solution: If they are both knights, then they will both say *"yes"*, and the same if they are both knaves. If one is a knight and the other a knave, then the knight would answer *"no"* truthfully and the knave would lie and also answer *"no"*. Therefore their answers are always the same.

1.12) Imagine the following rules for choosing optional subjects for the next academic year:

R1: If you choose *Computer Graphics*, then you must also choose *Numerical Programming* or *Algebra*.

R2: Choose *Algebra* if and only if you choose *Computer Graphics* or *Numerical Programming*.

R3: You cannot choose all three, but you must choose at least one of the three subjects.

R4: If you do not choose *Algebra*, then choose the other two.

(a) Is it possible to satisfy this set of rules?
Solution: Let p, q, r represent (respectively) choosing *Computer Graphics*, *Numerical Programming* or *Algebra*. then the rules above correspond to the following set of formulas:

$$\{p \rightarrow (q \vee r), (r \rightarrow (p \vee q)) \wedge ((p \vee q) \rightarrow r), (p \vee q \vee r) \wedge \neg(p \wedge q \wedge r), \neg r \rightarrow (p \wedge q)\}$$

This set is satisfiable for example by choosing *Algebra* and one of the other two.

(b) Is rule **R4** really necessary? What about rule **R1**?
Solution: **R4** it is not a semantic consequence of the remaining rules, therefore we cannot eliminate it and still impose the same rules. **R1** is a semantic consequence of the other rules, therefore we can remove it from the set of rules. This is a consequence of the following result: If $\Gamma \cup \{\varphi\} \models \psi$ and $\Gamma \models \varphi$, then $\Gamma \models \psi$.

(c) Give a simple rule that is equivalent to this (messy) set of four rules.
Solution: Having to choose at least one of the subjects is also a semantic consequence of the remaining rules. Therefore the following three rules are equivalent:

R1: Either choose *Algebra* and *Computer Graphics*, or *Algebra* and *Numerical Programming*.

R2: You cannot choose all three.

Al the rules above are semantic consequences of these rules, and vice-versa.

1.13) Justify the truth or falsity of each of the following statements where p, q, r and s are propositional variables.

(a) $\{(p \to q) \vee r, ((p \to q) \vee r) \to \neg r, (p \to q) \to (q \to r)\} \models \neg q$.
Solution: We will use truth tables. Note that we need to show that any assignment that satisfies all the formulas on the left of $\models$ also satisfies the formula on the right.

p	q	r	(p → q) ∨ r	((p → q) ∨ r) → ¬ r	(p → q) → (q → r)	¬ q
T	**T**	**T**	**T** **T**	**T** **T** **F** **F**	**T** **T** **T**	**F**
T	**T**	**F**	**T** **T**	**T** **T** **T** **T**	**T** **F** **F**	**F**
T	**F**	**T**	**F** **T**	**F** **T** **F** **F**	**F** **T** **T**	**T**
T	**F**	**F**	**F** **F**	**F** **F** **T** **T**	**F** **T** **T**	**T**
F	**T**	**T**	**T** **T**	**T** **T** **F** **F**	**T** **T** **T**	**F**
F	**T**	**F**	**T** **T**	**T** **T** **T** **T**	**T** **F** **F**	**F**
F	**F**	**T**	**T** **T**	**T** **T** **F** **F**	**T** **T** **T**	**T**
F	**F**	**F**	**T** **T**	**T** **T** **T** **T**	**T** **T** **T**	**T**

Note that, in the only line where the three formulas on the left are true, $\neg q$ is also true.

Another way to prove this statement is by contradiction. Let us assume that $\{(p \to q) \vee r, ((p \to q) \vee r) \to \neg r, (p \to q) \to (q \to r)\} \not\models \neg q$ For that to happen there is an assignment ν that satisfies $\{(p \to q) \vee r, ((p \to q) \vee r) \to \neg r, (p \to q) \to (q \to r)\}$ and does not satisfy $\neg q$, which means that it satisfies q. If ν satisfies q and $(p \to q) \to (q \to r)$ then ν satisfies r (otherwise $\not\models_\nu q \to r$, which would imply $\not\models_\nu (p \to q) \to (q \to r)$ since $\models_\nu p \to q$). Also, since ν satisfies r, then ν satisfies $(p \to q) \vee r$ and ν does not satisfy $\neg r$, which implies that ν does not satisfy $((p \to q) \vee r) \to \neg r$, leading to a contradiction. The contradiction follows from assuming there exists ν that satisfies $\{(p \to q) \vee r, ((p \to q) \vee r) \to \neg r, (p \to q) \to (q \to r)\}$ and does not satisfy $\neg q$, which means that this assignment does not exist, therefore $\{(p \to q) \vee r, ((p \to q) \vee r) \to \neg r, (p \to q) \to (q \to r)\} \models \neg q$.

(b) $\{(p \vee q) \to r\} \models (p \to r) \vee (q \to r)$.

Solution:

p	q	r	(p	∨	q)	→	r	(p	→	r)	∨	(q	→	r)
T	**T**	**T**		**T**		**T**			**T**		**T**		**T**	
T	**T**	**F**		**T**		**F**			**F**		**F**		**F**	
T	**F**	**T**		**T**		**T**			**T**		**T**		**T**	
T	**F**	**F**		**T**		**F**			**F**		**T**		**T**	
F	**T**	**T**		**T**		**T**			**T**		**T**		**T**	
F	**T**	**F**		**T**		**F**			**T**		**T**		**F**	
F	**F**	**T**		**F**		**T**			**T**		**T**		**T**	
F	**F**	**F**		**F**		**T**			**T**		**T**		**T**	

Again note that, all the assignments that satisfy $(p \vee q) \rightarrow r$ also satisfy $(p \rightarrow r) \vee (q \rightarrow r)$. Also note that reverse does not hold, therefore the two formulas are not equivalent. For $\nu(p) = \mathbf{T}$, $\nu(q) = \mathbf{F}$ and $\nu(r) = \mathbf{F}$, $\models_\nu (p \rightarrow r) \vee (q \rightarrow r)$ but $\not\models_\nu (p \vee q) \rightarrow r$.

1.14) Without building the truth table, justify the truth or falsity of each of the following statements where p, q, r and s are propositional variables.

(a) $\models (p \rightarrow (((q \rightarrow p) \rightarrow p) \rightarrow r)$.
Solution: For $\nu(p) = \mathbf{T}$ and $\nu(r) = \mathbf{F}$, we have $\models_\nu p$ and $\not\models_\nu ((q \rightarrow p) \rightarrow p) \rightarrow r$, therefore $\not\models_\nu (p \rightarrow (((q \rightarrow p) \rightarrow p) \rightarrow r))$.

(b) $\models (p \rightarrow (q \rightarrow s)) \rightarrow (q \rightarrow (p \rightarrow s))$.
Solution: This formula is a tautology. Let us first prove this by contradiction. We assume that the formula is not a tautology, that is, there exists an assignment ν that does not satisfy the formula. If $\not\models_\nu (p \rightarrow (q \rightarrow s)) \rightarrow (q \rightarrow (p \rightarrow s))$, then $\models_\nu p \rightarrow (q \rightarrow s)$ and $\not\models_\nu q \rightarrow (p \rightarrow s)$. From $\not\models_\nu q \rightarrow (p \rightarrow s)$ it follows that $\models_\nu q$ and $\not\models_\nu p \rightarrow s$, therefore $\models_\nu p$ and $\not\models_\nu s$. But from $\models_\nu q$, $\models_\nu p$ and $\not\models_\nu s$ it follows that $\not\models_\nu p \rightarrow (q \rightarrow s)$, which leads to a contradiction. This implies that the formula is a tautology.

One can also prove that a formula is a tautology using the laws of equivalence.

$$
\begin{aligned}
&(p \rightarrow (q \rightarrow s)) \rightarrow (q \rightarrow (p \rightarrow s)) \Leftrightarrow (\neg p \vee (q \rightarrow s)) \rightarrow (q \rightarrow (p \rightarrow s)) \\
&\Leftrightarrow (\neg p \vee (\neg q \vee s)) \rightarrow (q \rightarrow (p \rightarrow s)) \Leftrightarrow \neg(\neg p \vee (\neg q \vee s)) \vee (q \rightarrow (p \rightarrow s)) \\
&\Leftrightarrow \neg(\neg p \vee (\neg q \vee s)) \vee (\neg q \vee (p \rightarrow s)) \Leftrightarrow \neg(\neg p \vee (\neg q \vee s)) \vee (\neg q \vee (\neg p \vee s)) \\
&\Leftrightarrow (p \wedge (q \wedge \neg s)) \vee (\neg q \vee (\neg p \vee s)) \Leftrightarrow (p \wedge q \wedge \neg s) \vee \neg q \vee \neg p \vee s \\
&\Leftrightarrow ((p \vee \neg q) \wedge (q \vee \neg q) \wedge (\neg s \vee \neg q)) \vee \neg p \vee s \\
&\Leftrightarrow ((p \vee \neg q \vee \neg p) \wedge (q \vee \neg q \vee \neg p) \wedge (\neg s \vee \neg q \vee \neg p)) \vee s \\
&\Leftrightarrow (p \vee \neg q \vee \neg p \vee s) \wedge (q \vee \neg q \vee \neg p \vee s) \wedge (\neg s \vee \neg q \vee \neg p \vee s)
\end{aligned}
$$

Recall that a formula in conjunctive normal form is a tautology if all disjunctions are tautologies, which is the case of this formula because all disjuntions have a literal and its complement: the first one has p and $\neg p$, the second q and $\neg q$ and the third one s and $\neg s$. Note that if, in addition to the laws of equivalence given in the course we consider the following laws for neutral and absorbing elements, we can effectively show that the formula is equivalent to $\mathbf{T}$.

$$
\begin{aligned}
(\phi \vee \mathbf{F}) &\Leftrightarrow \phi \quad \text{(neutral element of } \vee\text{)} \\
(\phi \wedge \mathbf{T}) &\Leftrightarrow \phi \quad \text{(neutral element of } \wedge\text{)} \\
(\phi \vee \mathbf{T}) &\Leftrightarrow \mathbf{T} \quad \text{(absorbing element of } \vee\text{)} \\
(\phi \wedge \mathbf{F}) &\Leftrightarrow \mathbf{F} \quad \text{(absorbing element of } \wedge\text{)}
\end{aligned}
$$

(c) $\models (p \to (q \to r)) \to ((p \to q) \to (p \to r))$.
Solution: Let us proceed by contradiction. We assume that $\not\models (p \to (q \to r)) \to ((p \to q) \to (p \to r))$, therefore there exists an assignment such that $\not\models_\nu (p \to (q \to r)) \to ((p \to q) \to (p \to r))$, which implies $\models_\nu p \to (q \to r)$ and $\not\models_\nu (p \to q) \to (p \to r)$. But $\not\models_\nu (p \to q) \to (p \to r)$ implies $\not\models_\nu p \to r$, therefore $\models_\nu p$ and $\not\models_\nu r$. But then, $\models_\nu p \to (q \to r)$ implies that $\not\models_\nu q$, which then implies that $\not\models_\nu p \to q$, which leads to $\models_\nu (p \to q) \to (p \to r)$, contradicting what was previously stated. Therefore $\models (p \to (q \to r)) \to ((p \to q) \to (p \to r))$.

(d) $\models ((p \wedge q) \to (s \vee t)) \to ((p \to s) \vee (q \to t))$.
Solution: In this case let us use semantic equivalence.

$$
\begin{aligned}
&((p \wedge q) \to (s \vee t)) \to ((p \to s) \vee (q \to t)) \Leftrightarrow \neg((p \wedge q) \to (s \vee t)) \vee ((p \to s) \vee (q \to t)) \\
&\Leftrightarrow \neg((p \wedge q) \to (s \vee t)) \vee ((p \to s) \vee (q \to t)) \Leftrightarrow \neg(\neg(p \wedge q) \vee (s \vee t)) \vee ((p \to s) \vee (q \to t)) \\
&\Leftrightarrow \neg(\neg(p \wedge q) \vee (s \vee t)) \vee ((\neg p \vee s) \vee (\neg q \vee t)) \Leftrightarrow (\neg\neg(p \wedge q) \wedge \neg(s \vee t)) \vee (\neg p \vee s \vee \neg q \vee t) \\
&\Leftrightarrow (p \wedge q \wedge \neg s \wedge \neg t) \vee \neg p \vee s \vee \neg q \vee t \\
&\Leftrightarrow ((p \vee \neg p) \wedge (q \vee \neg p) \wedge (\neg s \vee \neg p) \wedge (\neg t \vee \neg p)) \vee s \vee \neg q \vee t \\
&\Leftrightarrow ((p \vee \neg p \vee s) \wedge (q \vee \neg p \vee s) \wedge (\neg s \vee \neg p \vee s) \wedge (\neg t \vee \neg p \vee s)) \vee \neg q \vee t \\
&\Leftrightarrow ((p \vee \neg p \vee s \vee \neg q) \wedge (q \vee \neg p \vee s \vee \neg q) \wedge (\neg s \vee \neg p \vee s \vee \neg q) \wedge (\neg t \vee \neg p \vee s \vee \neg q)) \vee t \\
&\Leftrightarrow (p \vee \neg p \vee s \vee \neg q \vee t) \wedge (q \vee \neg p \vee s \vee \neg q \vee t) \wedge (\neg s \vee \neg p \vee s \vee \neg q \vee t) \wedge (\neg t \vee \neg p \vee s \vee \neg q \vee t)
\end{aligned}
$$

Again, the formula is a tautology since all its disjunctions are tautologies.

1.15) Justify the truth or falsity of each of the following statements, where Γ and Σ represent sets of formulas and $\phi, \psi, \theta, \gamma$ represent formulas of propositional logic. If the statement is false, you must give a counterexample indicating particular values of the sets or formulas for which the statement is false. If the statement is true you must justify using the definitions.

(a) $\models (\theta \vee \phi) \to (\neg\theta \to \phi)$.
Solution:

$$
\begin{aligned}
(\theta \vee \phi) \to (\neg\theta \to \phi) &\Leftrightarrow \neg(\theta \vee \phi) \vee (\neg\theta \to \phi) \\
&\Leftrightarrow \neg(\theta \vee \phi) \vee (\neg\neg\theta \vee \phi) \\
&\Leftrightarrow (\neg\theta \wedge \neg\phi) \vee (\theta \vee \phi) \\
&\Leftrightarrow (\neg\theta \vee \theta \vee \phi) \wedge (\neg\phi \vee \theta \vee \phi) \\
&\Leftrightarrow (\mathbf{T} \vee \phi) \wedge (\mathbf{T} \vee \theta) \\
&\Leftrightarrow \mathbf{T} \wedge \mathbf{T} \Leftrightarrow \mathbf{T}
\end{aligned}
$$

(b) $\{\phi \to \psi, \psi \to \phi\} \models (\phi \vee \psi) \to (\phi \wedge \psi)$.

ϕ	ψ	$\phi \to \psi$	$\psi \to \phi$	$(\phi \vee \psi)$	$\to$	$(\phi \wedge \psi)$
T	**T**	**T**	**T**	**T**	**T**	**T**
T	**F**	**F**	**T**	**T**	**F**	**F**
F	**T**	**T**	**F**	**T**	**F**	**F**
F	**F**	**T**	**T**	**F**	**T**	**F**

Note than in every line where the two formulas $\phi \to \psi$, $\psi \to \phi$ are simultaneously true, the consequence is also true. Therefore the semantic consequence holds.

(c) $\{\psi \to (\neg\phi \vee \gamma), \neg\gamma\} \models \neg\phi \to \neg\psi$.
Solution: Consider ν such that $\nu(\phi) = \mathbf{F}$, $\nu(\psi) = \mathbf{T}$ and $\nu(\gamma) = \mathbf{F}$. Then we have $\models_\nu \{\psi \to (\neg\phi \vee \gamma), \neg\gamma\}$, but $\not\models_\nu \neg\phi \to \neg\psi$, therefore $\{\psi \to (\neg\phi \vee \gamma), \neg\gamma\} \not\models \neg\phi \to \neg\psi$

(d) If $\phi, \psi \models \theta \to \gamma$ and $\phi \models \theta$ then $\phi, \psi \models \gamma$.
Solution: Let us assume by contradiction that $\phi, \psi \models \theta \to \gamma$, $\phi \models \theta$ but $\phi, \psi \not\models \gamma$. This means

that there is an assignment ν, such that $\models_\nu \phi$, $\models_\nu \psi$ and $\not\models_\nu \gamma$. But then, from that and the fact that $\phi, \psi \models \theta \rightarrow \gamma$ we have $\not\models_\nu \theta$, which contradicts $\phi \models \theta$. Therefore the implication is true.

1.16) Justify the truth or falsity of each of the following statements, where Γ and Σ represent sets of formulas and $\phi, \psi, \theta, \gamma$ represent formulas of propositional logic. If the statement is false, you must give a counterexample indicating particular values of the sets or formulas for which the statement is false. If the statement is true you must justify using the definitions.

(a) If $\Gamma \models \theta$ and $\Gamma \subseteq \Sigma$, then $\Sigma \models \theta$.
Solution: The statement is true. To prove, let us assume that $\Gamma \models \theta$ and $\Gamma \subseteq \Sigma$ and we want to prove that $\Sigma \models \theta$. To prove that $\Sigma \models \theta$ we need to show that any assignment ν that satisfies all the formulas in Σ also satisfies θ (if $\models_\nu \Sigma$ then $\models_\nu \theta$). Let ν be any assignment that satisfies Σ. Since all the formulas in Γ are a part of Σ then ν satisfies Γ, but $\Gamma \models \theta$, therefore any assignment that satisfies Γ also satisfies θ, which implies that ν satisfies θ.

(b) If $\Sigma \models \theta$ and $\Gamma \subseteq \Sigma$, then $\Gamma \models \theta$.
Solution: The statement is false. Let $\Sigma = \{p, p \rightarrow q\}$, $\theta = q$ and $\Gamma = \{p\}$. It is easy to see that $\{p, p \rightarrow q\} \models q$ and $\{p\} \subseteq \{p, p \rightarrow q\}$, but $\{p\} \not\models q$.

(c) If $\Gamma \models \theta$ and $\Sigma \models \theta$, then $\Gamma \cap \Sigma \models \theta$.
Solution: The statement is false. Let $\Sigma = \{p, p \rightarrow q\}$, $\Gamma = \{q\}$ and $\theta = q$. It is easy to see that $\{p, p \rightarrow q\} \models q$ and $\{q\} \models q$ but $\{p\} \cap \{p, p \rightarrow q\} = \varnothing$ and $\not\models q$ (q is not a tautology).

(d) If $\Gamma \models \theta$ and $\Sigma \subseteq \Gamma$ then $\Sigma \cap \Gamma \models \theta$.
Solution: The statement is false. Again let $\Gamma = \{p, p \rightarrow q\}$, $\theta = q$ and $\Sigma = \{p\}$. It is easy to see that $\{p, p \rightarrow q\} \models q$ and $\{p\} \subseteq \{p, p \rightarrow q\}$, but $\{p\} \cap \{p, p \rightarrow q\} = \{p\}$ and $\{p\} \not\models q$.

(e) If $\Sigma \models \theta$ then $\Sigma \cap \Gamma \models \theta$. **Solution:** The statement is false. Let $\Sigma = \{p\}$, $\theta = p$ and $\Gamma = \varnothing$. Then it is easy to see that $\{p\} \models p$ but $\{p\} \cap \varnothing = \varnothing$ and $\not\models p$ (p is not a tautology).

(f) If $\Sigma \models \theta$ and $\Sigma \subseteq \Gamma$ then $\Sigma \cup \Gamma \models \theta$.
Solution: The statement is true. Note that, if $\Sigma \subseteq \Gamma$ then $\Sigma \cup \Gamma = \Gamma$, therefore the result follows from (a).

1.17) For each of the truth functions (a), (b) and (c), construct a formula in disjunctive normal form and a formula in conjunctive normal form:

p	q	r	(a)	(b)	(c)
T	**T**	**T**	**T**	**F**	**T**
T	**T**	**F**	**T**	**T**	**F**
T	**F**	**T**	**T**	**F**	**T**
T	**F**	**F**	**T**	**T**	**F**
F	**T**	**T**	**F**	**T**	**T**
F	**T**	**F**	**T**	**T**	**F**
F	**F**	**T**	**T**	**F**	**T**
F	**F**	**F**	**T**	**F**	**F**

Solution: We will used the method studied in the course to define CNFs and DNFs using the truth table.

(a) Disjunctive normal form:

$$(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$$

Conjunctive normal form:

$$(p \vee \neg q \vee \neg r)$$

(b) Disjunctive normal form:

$$(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r)$$

Conjunctive normal form:

$$(\neg p \vee \neg q \vee \neg r) \wedge (\neg p \vee q \vee \neg r) \wedge (p \vee q \vee \neg r) \wedge (p \vee q \vee r)$$

(c) Disjunctive normal form:

$$(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$$

Conjunctive normal form:

$$(\neg p \vee \neg q \vee r) \wedge (\neg p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee q \vee r)$$

1.19) Sets of connectives

(a) Show that the logical connectives $\{\neg, \vee\}$ are a universal (complete) set. That is, propositions of the form $p \rightarrow q$, $p \leftrightarrow q$, and $p \wedge q$ can be written using only the connectives $\neg$ and $\vee$.
Solution: First we show that $\rightarrow$ is not needed since we have $p \rightarrow q \Leftrightarrow \neg p \vee q$. We have also seen in lectures that $p \leftrightarrow q \Leftrightarrow (p \wedge q) \vee (\neg p \wedge \neg q)$. Finally, $(p \wedge q) \Leftrightarrow \neg(\neg p \vee \neg q)$. Which shows that any statement using these connectives can be written using just $\{\neg, \vee\}$.

Note that we have proved in this course that $\{\neg, \wedge, \vee\}$ is a complete set of connectives. So any other complete set of connectives just needs to be able to express any formula involving the connectives in $\{\neg, \wedge, \vee\}$.

(b) Define a new connective $\overline{\wedge}$ as follows:

p	q	$p \overline{\wedge} q$
T	**T**	**F**
T	**F**	**T**
F	**T**	**T**
F	**F**	**T**

Show that the propositions $\neg p$, $p \wedge q$, and $p \vee q$ can be written using only the connective $\overline{\wedge}$. Conclude that $\overline{\wedge}$ is universal (complete).
Solution: We show that $\neg$ is not necessary since $\neg p \Leftrightarrow p \overline{\wedge} p$. We can eliminate $\wedge$, since $p \wedge q \Leftrightarrow \neg(p \overline{\wedge} q) \Leftrightarrow (p \overline{\wedge} q) \overline{\wedge} (p \overline{\wedge} q)$.

p	$p \overline{\wedge} p$
T	**F**
F	**T**

p	q	$\neg$	$(p \overline{\wedge} q)$	$(p \overline{\wedge} q) \overline{\wedge} (p \overline{\wedge} q)$
T	**T**	**T**	**F**	**F**
T	**F**	**F**	**T**	**T**
F	**T**	**F**	**T**	**T**
F	**F**	**F**	**T**	**T**

And $p \vee q \Leftrightarrow \neg(\neg p \wedge \neg q)$, which we have already seen how we can write in terms of $\overline{\wedge}$. So $\{\overline{\wedge}\}$ is complete.

1.21) Consider the satisfiability algorithm for Horn's formulas given in the course.

(c) Apply the algorithm to the following formulas:

- $(\neg p \vee \neg q) \wedge (\neg q \vee r) \wedge q$
  Solution: We start by checking if any of the propositional variables is one of the elements of the conjunction and propagate that information in the formula.

p	q	r	$(\neg p \vee \neg q) \wedge (\neg q \vee r) \wedge q$
	T		**F** **F**

  Given this, other values are determined for the propositional variables p and r:

p	q	r	$(\neg p \vee \neg q) \wedge (\neg q \vee r) \wedge q$
F	**T**	**T**	**F** **F**

  Since all the variables are assigned and no disjunction was assigned **F** then the algorithm terminates.

- $p \wedge (\neg p \vee q) \wedge (\neg q \vee p)$
  Solution: In this case we start by assigning the value **T** to p:

p	q	r	$p \wedge (\neg p \vee q) \wedge (\neg q \vee p)$
T			**F** **T**

  This determines the following assignment for q:

p	q	r	$p \wedge (\neg p \vee q) \wedge (\neg q \vee p)$
T	**T**		**F** **T** **F** **T**

  Since we have not assigned **F** to any disjunction and we cannot propagate more this information, then we can safely assign **F** to the remaining variables and **T** to the formula:

p	q	r	$p \wedge (\neg p \vee q) \wedge (\neg q \vee p)$
T	**T**	**F**	**F** **T** **F** **T**

- $\neg p \wedge (\neg p \vee q) \wedge \neg q$
  Solution: Since no variable is an element of the conjunction then we can safely satisfy the formula by assigning **F** all the variables:

p	q	r	$\neg p \wedge (\neg p \vee q) \wedge \neg q$
F	**F**	**F**	

- $p \wedge (\neg p \vee q) \wedge \neg r$
  Solution: In this case we start by assigning the value **T** to p:

p	q	r	$p \wedge (\neg p \vee q) \wedge \neg r$
T			**F**

  This determines the following assignment for q:

p	q	r	$p \wedge (\neg p \vee q) \wedge \neg r$
T	**T**		**F** **T**

  Since we have not assigned **F** to any disjunction and we cannot propagate more this information, then we can safely assign **F** to the remaining variables and **T** to the formula:

p	q	r	$p \wedge (\neg p \vee q) \wedge \neg r$
T	**T**	**F**	**F** **T**

1.22) Apply the David-Putman (DLL) algorithm to the following sets of clauses.

(a)

$$\{p \vee q \vee r, \neg p \vee \neg q \vee \neg r, p \vee \neg q \vee \neg r, p \vee q \vee \neg r, \neg p \vee q, \neg p \vee r, p \vee \neg q \vee r\}$$

Solution: Let
$S = \{p \vee q \vee r, \neg p \vee \neg q \vee \neg r, p \vee \neg q \vee \neg r, p \vee q \vee \neg r, \neg p \vee q, \neg p \vee r, p \vee \neg q \vee r\}$
Since there are no unit-clauses we need to select a literal. Let us choose p and call DLL($S \cup \{p\}$).
Applying unit-propagation for p to $S \cup \{p\}$ we obtain:
$S_1 = \{\neg q \vee \neg r, q, r\}$
Applying unit-propagation for q to S_1 we obtain:
$S_2 = \{\neg r, r\}$
Applying unit-propagation for r to S_2, we obtain $\{\mathbf{F}\}$.

Going back to the last point where a choice was made, we select the complementary literal and call DLL($S \cup \{\neg p\}$). Applying unit-propagation for $\neg p$ to $S \cup \{\neg p\}$ we obtain:
$S_3 = \{q \vee r, \neg q \vee \neg r, q \vee \neg r, \neg q \vee r\}$
Since there are no unit-clauses we need to select a literal. Let us choose q and call DLL($S_3 \cup \{q\}$).
Applying unit-propagation for q to $S_3 \cup \{q\}$ we obtain:
$S_4 = \{\neg r, r\}$
Applying unit-propagation for r to S_4, we obtain $\{\mathbf{F}\}$.

Therefore the initial set of clauses is not satisfiable.

(b)

$$\{\neg p \vee q \vee r, p \vee q \vee r, p \vee q \vee \neg r, p \vee \neg q \vee r, p \vee \neg q \vee \neg r\}$$

Solution: Let
$S = \{\neg p \vee q \vee r, p \vee q \vee r, p \vee q \vee \neg r, p \vee \neg q \vee r, p \vee \neg q \vee \neg r\}$
Since there are no unit-clauses we need to select a literal. Let us choose p and call DLL($S \cup \{p\}$).
Applying unit-propagation for p to $S \cup \{p\}$ we obtain:
$S_1 = \{q \vee r\}$
Since there are no unit-clauses we need to select a literal. Let us choose q and call DLL($S_1 \cup \{q\}$).
Applying unit-propagation for q to S_1, we obtain $\{\}$.
Therefore the initial set of clauses is satisfiable for $\nu(p) = \mathbf{T}$ and $\nu(q) = \mathbf{T}$.

(c)

$$\{p \vee \neg q \vee \neg r, \neg p \vee q \vee r, \neg p \vee q, \neg p \vee r\}$$

Let
$S = \{p \vee \neg q \vee \neg r, \neg p \vee q \vee r, \neg p \vee q, \neg p \vee r\}$
Since there are no unit-clauses we need to select a literal. Let us choose p and call DLL($S \cup \{\neg p\}$).
Applying unit-propagation for $\neg p$ to $S \cup \{\neg p\}$ we obtain:
$S_1 = \{\neg q \vee \neg r\}$
Since there are no unit-clauses we need to select a literal. Let us choose q and call DLL($S_1 \cup \{\neg q\}$).
Applying unit-propagation for $\neg q$ to S_1, we obtain $\{\}$.
Therefore the initial set of clauses is satisfiable for $\nu(p) = \mathbf{F}$ and $\nu(q) = \mathbf{F}$.

(d)

$$\{p \vee \neg q \vee r, \neg p \vee q \vee r, p \vee \neg q \vee \neg r, \neg p \vee q \vee \neg r, \neg p \vee \neg q \vee \neg r\}$$

Let
$S = \{p \vee \neg q \vee r, \neg p \vee q \vee r, p \vee \neg q \vee \neg r, \neg p \vee q \vee \neg r, \neg p \vee \neg q \vee \neg r\}$
Since there are no unit-clauses we need to select a literal. Let us choose p and call $\mathrm{DLL}(S \cup \{\neg p\})$.
Applying unit-propagation for $\neg p$ to $S \cup \{\neg p\}$ we obtain:
$S_1 = \{\neg q \vee r, \neg q \vee \neg r\}$
Since there are no unit-clauses we need to select a literal. Let us choose q and call $\mathrm{DLL}(S_1 \cup \{\neg q\})$.
Applying unit-propagation for $\neg q$ to S_1, we obtain $\{\}$.
Therefore the initial set of clauses is satisfiable for $\nu(p) = \mathbf{F}$ and $\nu(q) = \mathbf{F}$.

(e)

$$\{p \vee \neg q \vee \neg r, \neg p \vee q \vee r, \neg p \vee \neg q \vee \neg r, \neg p \vee \neg q \vee r, \neg p \vee q \vee \neg r, p \vee q \vee \neg r\}$$

Let
$S = \{p \vee \neg q \vee \neg r, \neg p \vee q \vee r, \neg p \vee \neg q \vee \neg r, \neg p \vee \neg q \vee r, \neg p \vee q \vee \neg r, p \vee q \vee \neg r\}$
Since there are no unit-clauses we need to select a literal. Let us choose p and call $\mathrm{DLL}(S \cup \{\neg p\})$.
Applying unit-propagation for $\neg p$ to $S \cup \{\neg p\}$ we obtain:
$S_1 = \{\neg q \vee \neg r, q \vee \neg r\}$
Since there are no unit-clauses we need to select a literal. Let us choose $\neg r$ and call $\mathrm{DLL}(S_1 \cup \{\neg r\})$.
Applying unit-propagation for $\neg r$ to S_1, we obtain $\{\}$.
Therefore the initial set of clauses is satisfiable for $\nu(p) = \mathbf{F}$ and $\nu(r) = \mathbf{F}$.